

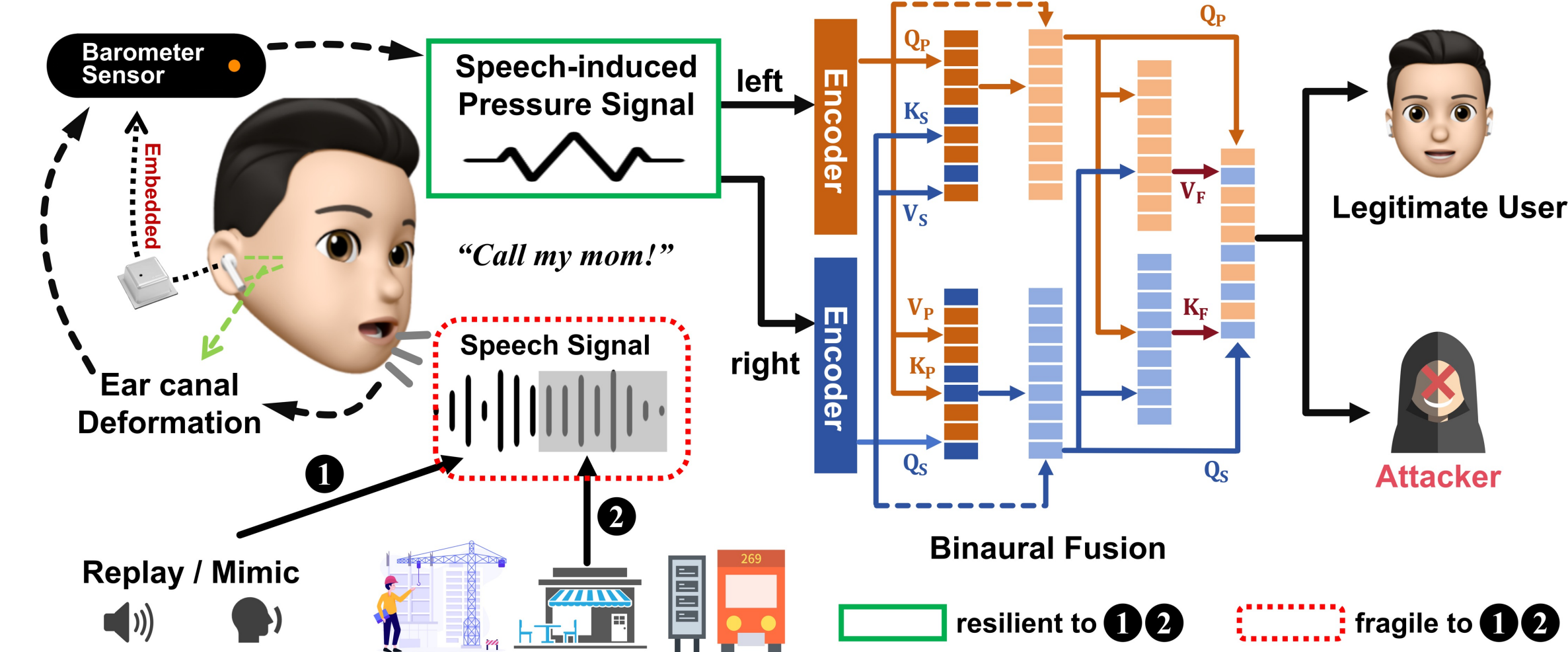
BaroAuth: Harnessing Ear Canal Deformation for Speaking User Authentication on Earbuds

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1. ABSTRACT & MOTIVATION

Problem: Current earable authentication schemes suffers from noise, spoofing, and high power cost.



Solution: we propose BaroAuth, a novel system using Speech-aware Pressure Sequences (SPSs).

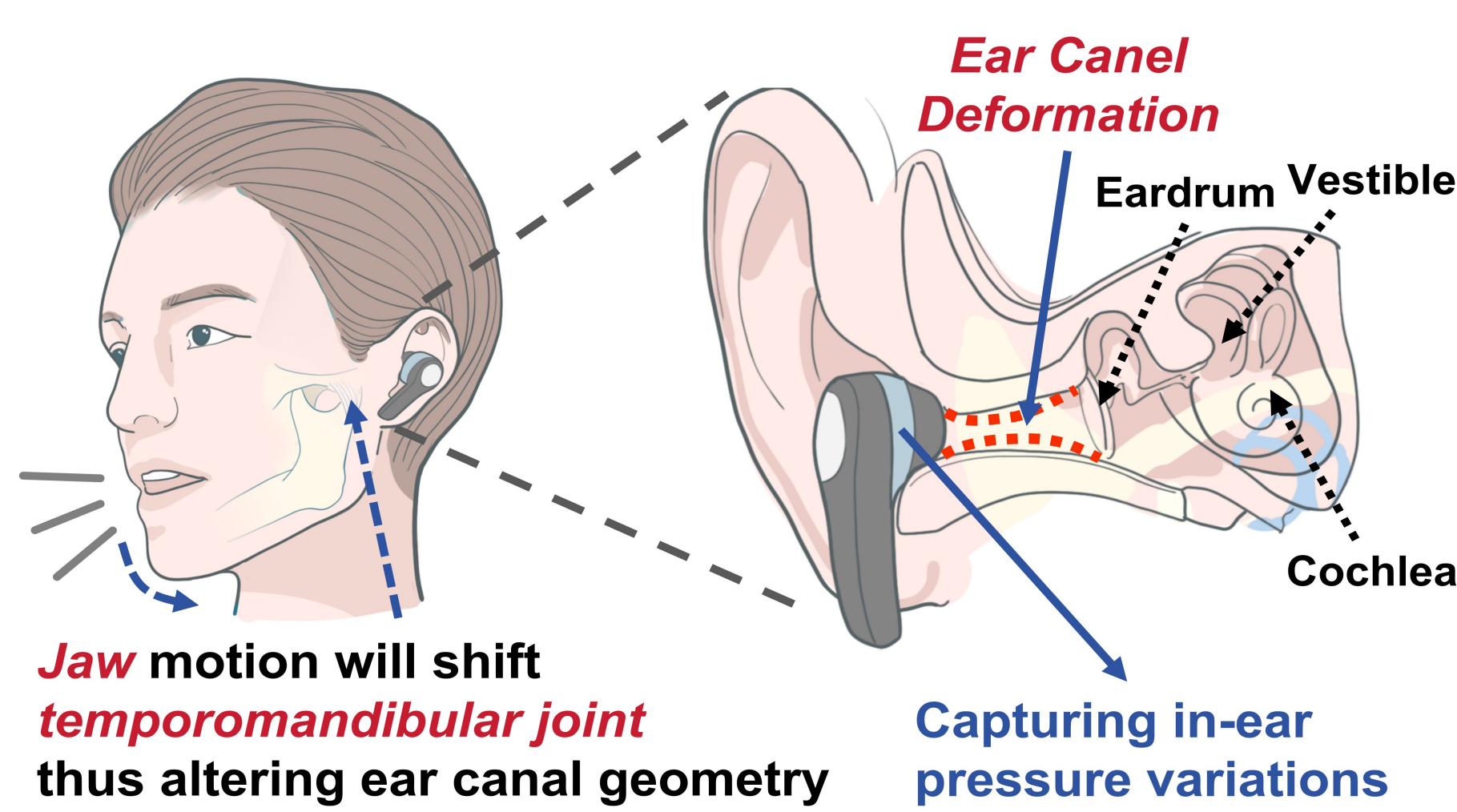
2. SENSING PRINCIPLE (SPS)

PHYSIOLOGICAL

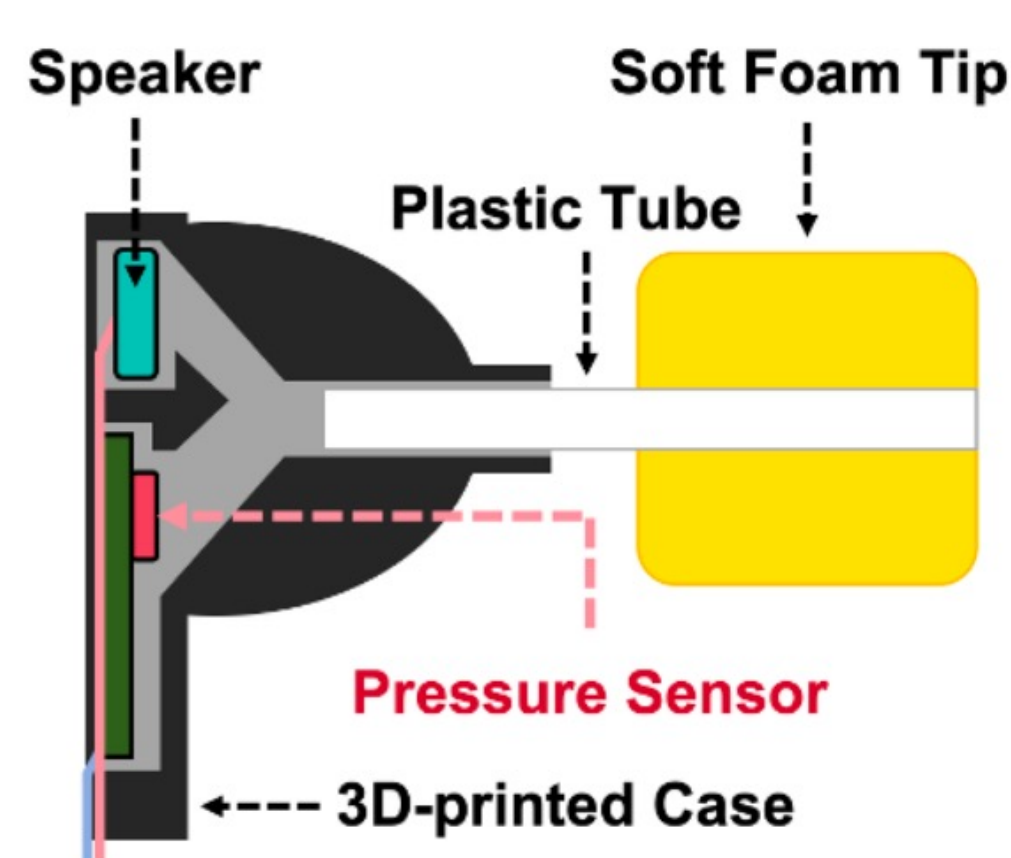
Ear canal geometry & TMJ structure unique to user.

PHYSIOLOGICAL

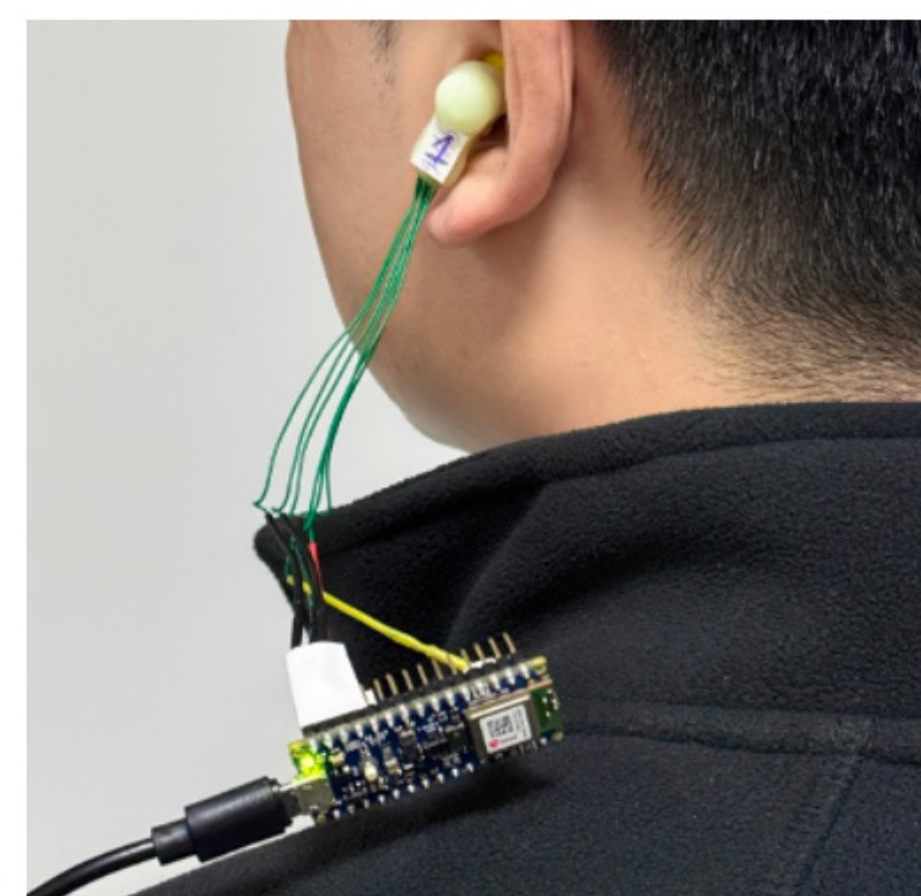
Ear canal geometry & TMJ structure unique to user.



3. DATA COLLECTION



(a) Hardware Schematic



(b) Data-collection Setup

Trace-A

60 day
Legal SPSs

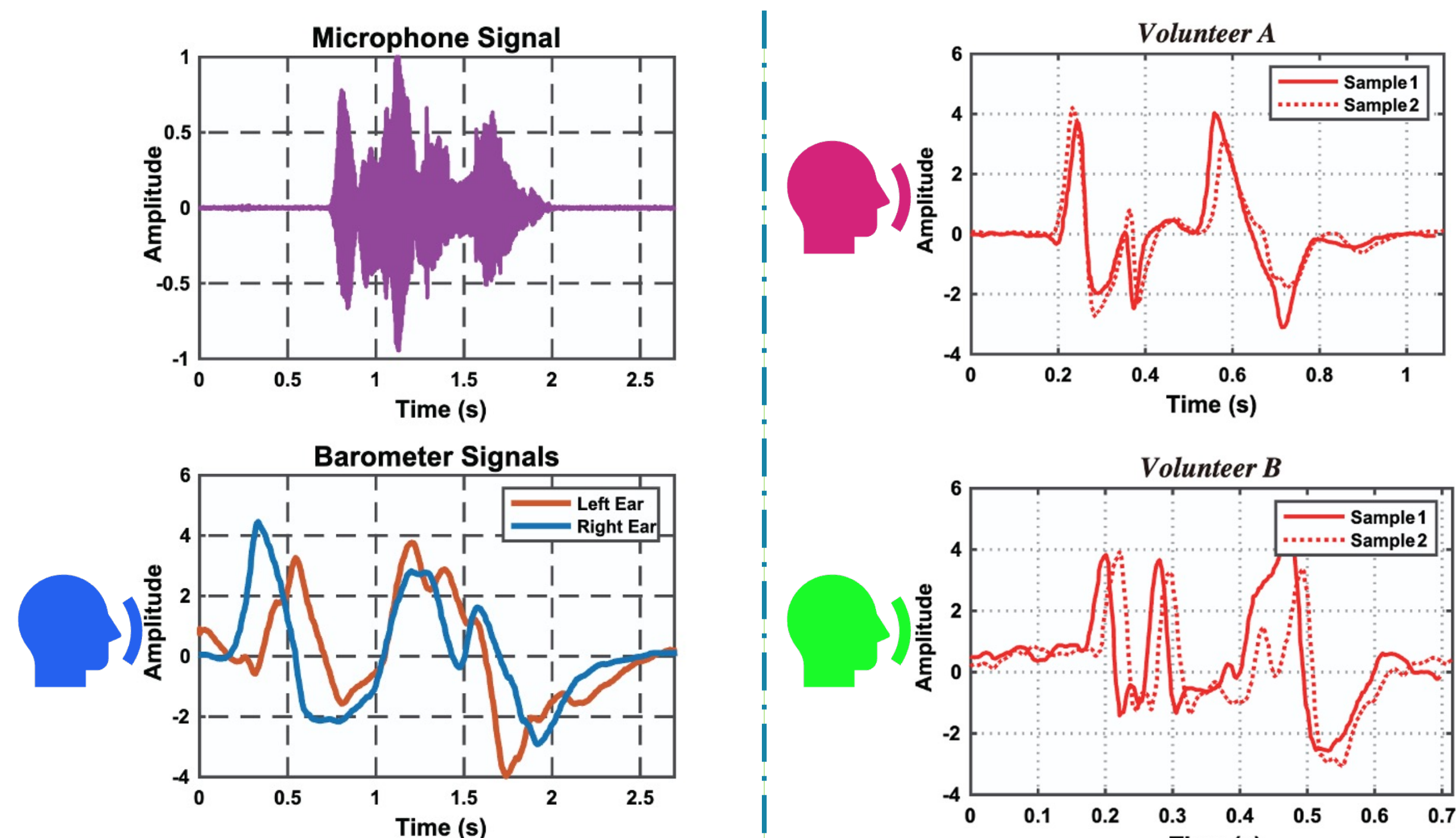
Trace-B

100 x
Spoofing Attacks

Trace-C

1 week
Long-term Stability

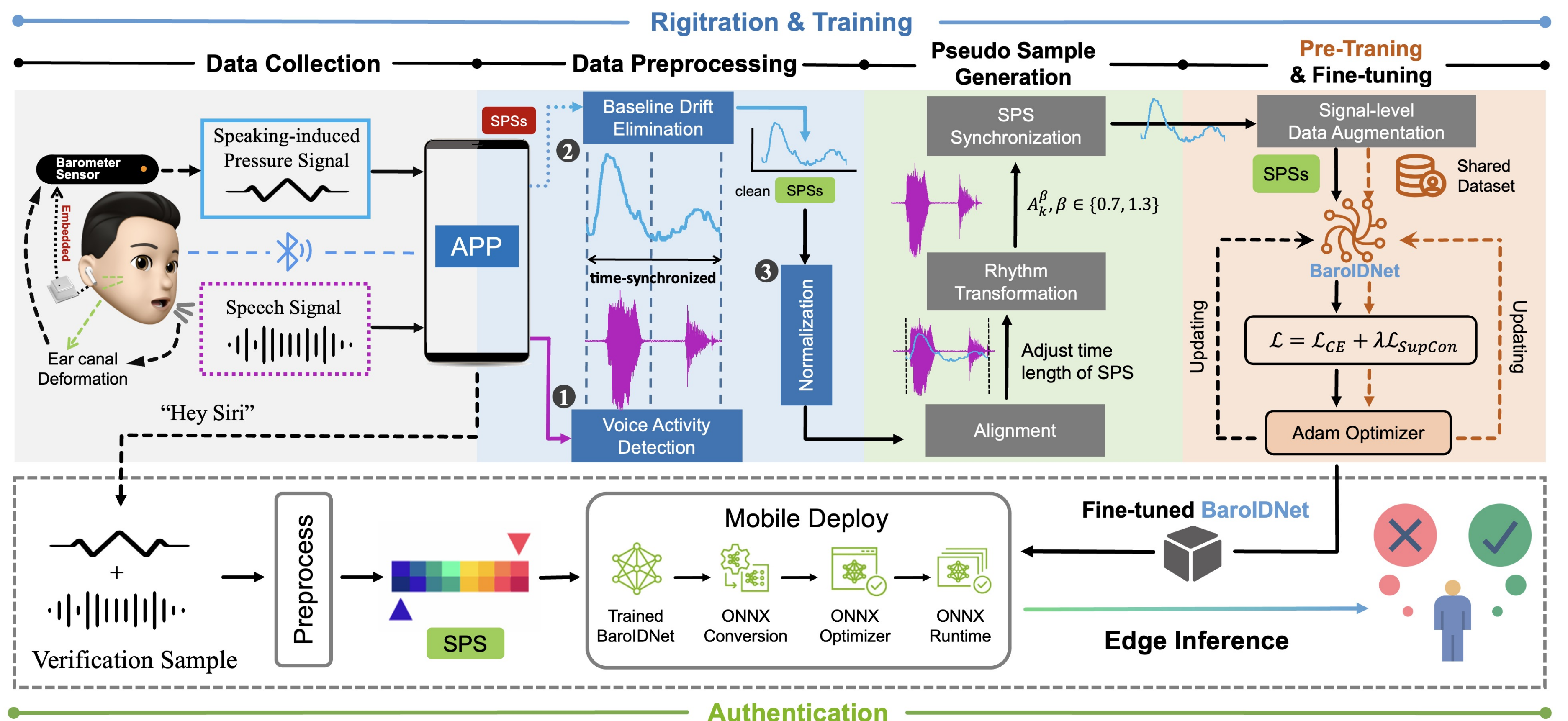
4. KEY OBSERVATION



✓ **Detectable**

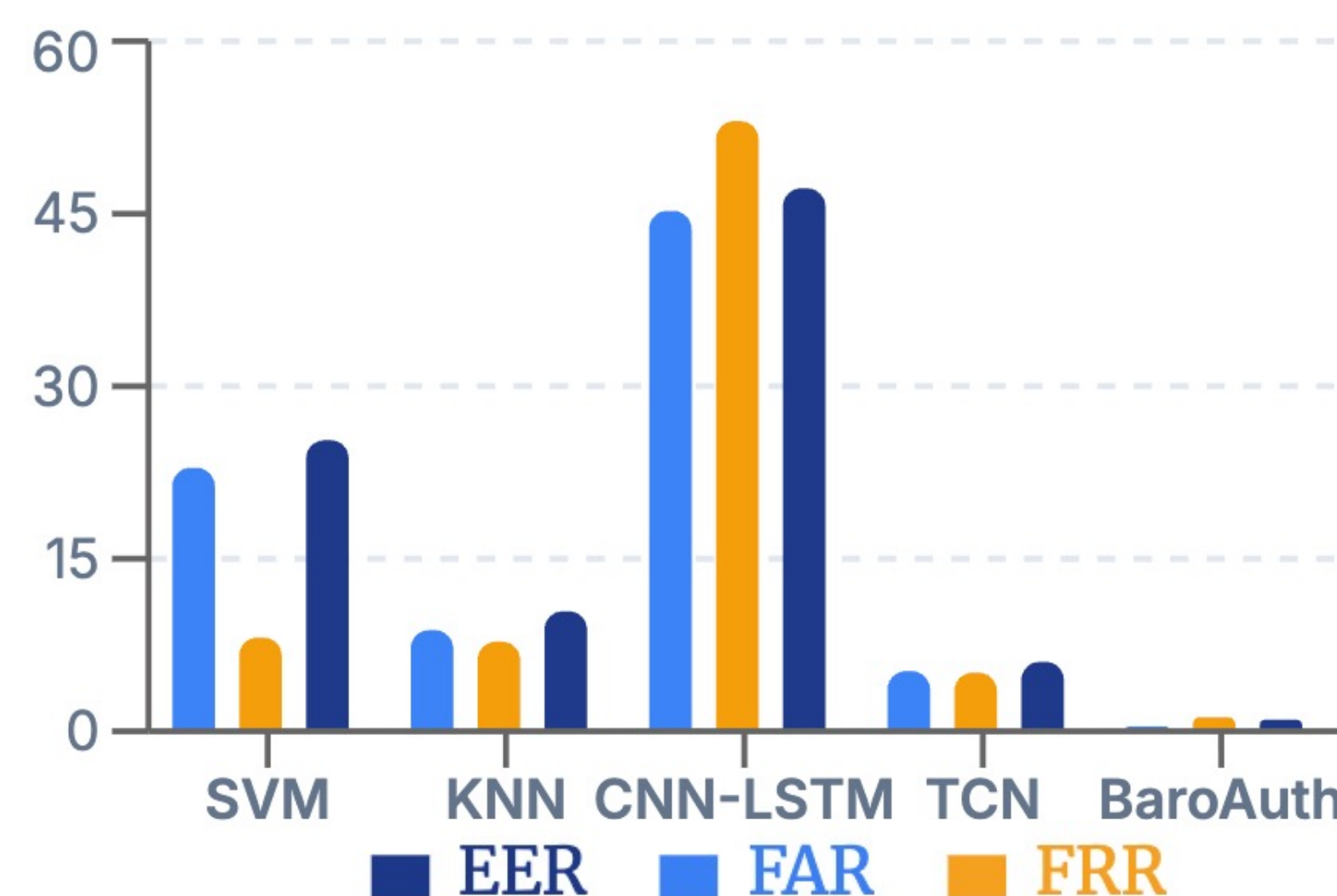
✗ **Different**

5. OVERVIEW of BaroAuth



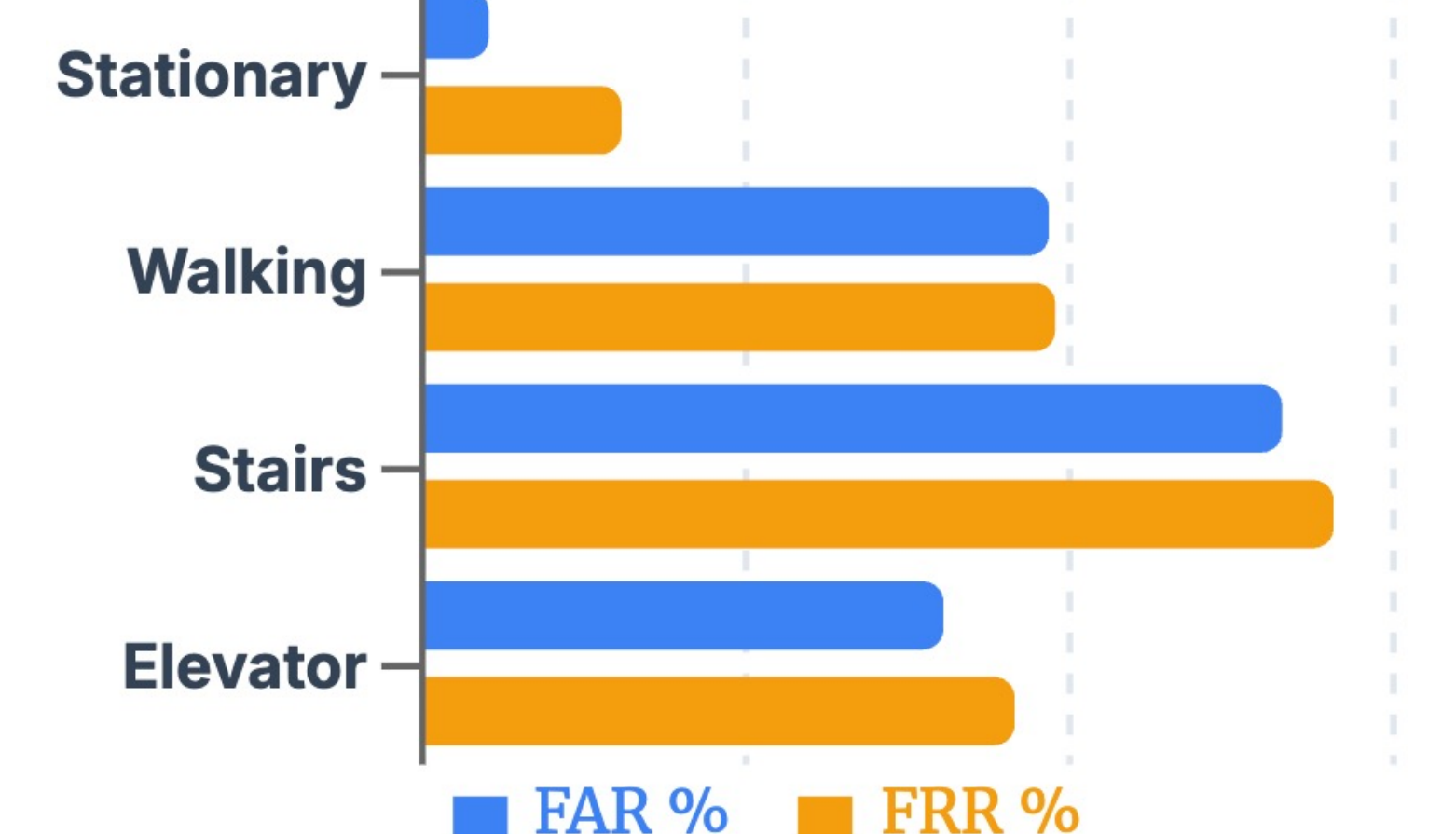
6. PERFORMANCE EVALUATION

PERFORMANCE COMPARISON



BaroAuth achieves the best 1.02% across all baselines.

MOBILE SCENARIOS



Stable across daily activities. EMD keeps motion error low (~3-5%)

RHYTHM HANDLING (SLOW/FAST)

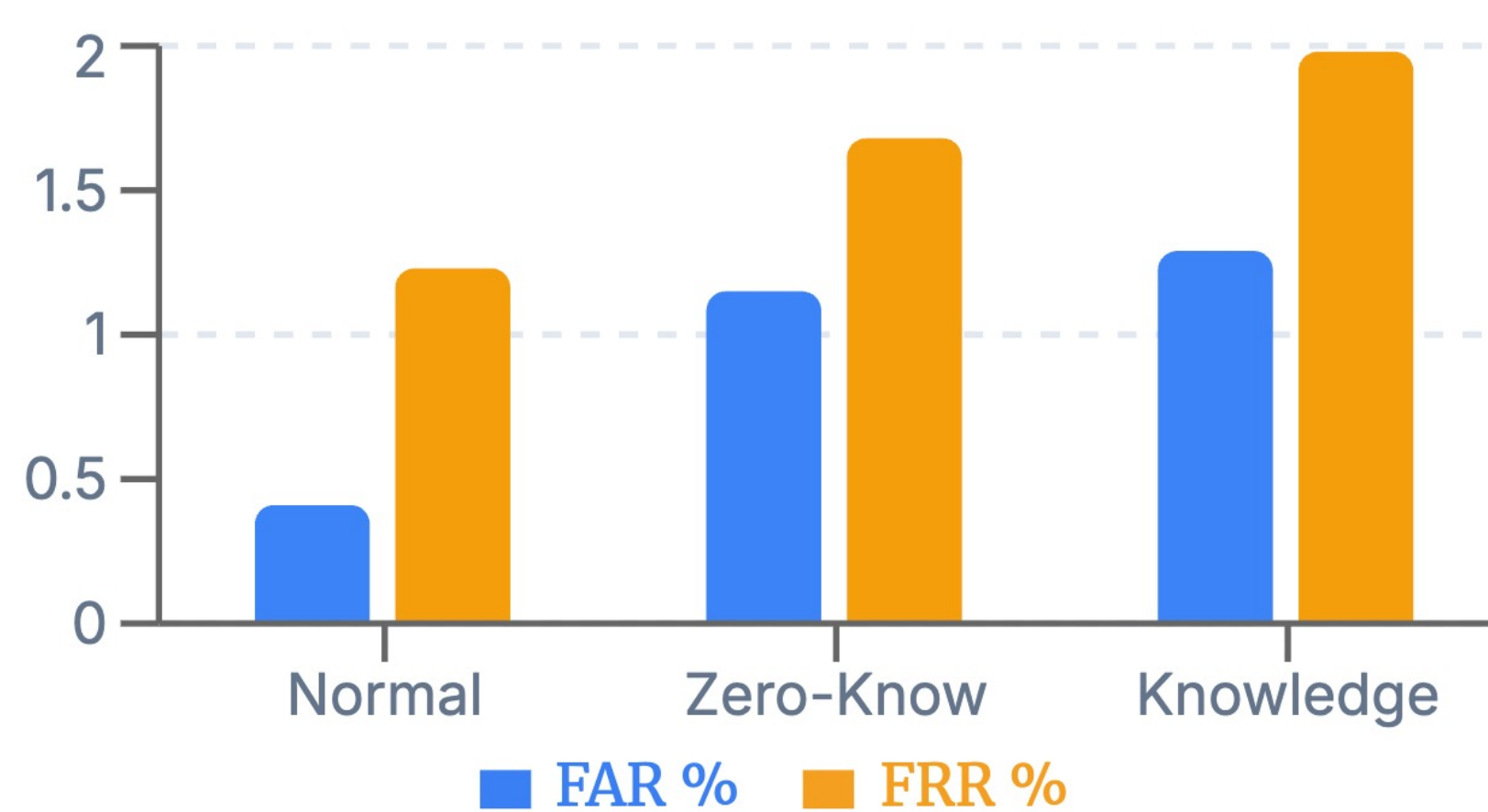


Rhythm Robustness

Without adaption (Red), fast/slow speech spikes error rates (>7%). Our Pseudo Sample Generation (Green) reduces error to <2% proving the model learns underlying biometric traits, not just tempo.

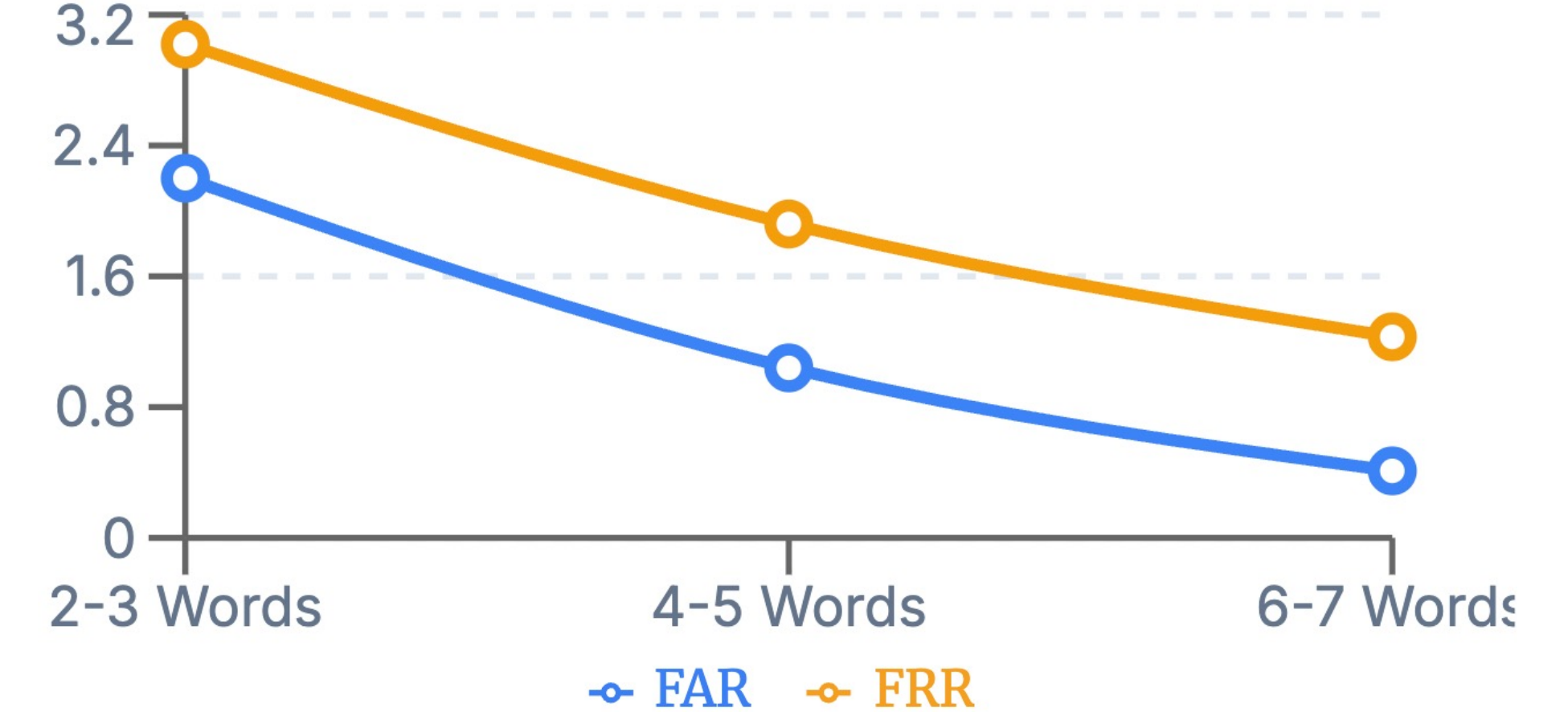
7. SECURITY & USABILITY

ATTACK RESILIENCE



Knowledge Attack: Even attackers mimic the user's speaking style visually, they cannot replicate internal ear pressure.

COMMAND LENGTH IMPACT



Trade-off: Longer commands (6-7 words) provide more phonetic data, reducing FAR to 0.41%.

8. CONCLUSION

BaroAuth pioneers passive, privacy authentication for earables.

- High Accuracy (99%+) & Low Power (13 mW).
- Immune to environmental noise (unlike voice).
- Robust against sophisticated mimicry attacks.

0.41%
FALSE ACCEPT RATE

18 Hz
LOW SAMPLING RATE